A quadratic hazard potential and a one-sample matroid prophet guarantee above 2^{-36}

Ryan Shea

October 10, 2026

Abstract

A quadratic query potential yields expected exit hazard at least $1/2$ in the reverse computation of the OpenAI matroid prophet construction. This constant is sharp under the abstract query hypotheses. Integrating the residual tail, changing the filter and density probabilities, and refining the finite rounding phases gives a proposed guarantee above 2^{-36} with $38n + 23$ initial fair bits, against an adversary seeing all samples, values, and the full seed. A compact exact coefficient is $1.0405020\ldots\ 2^{-36}$. This AI-assisted work has a written proof and same-agent verification, including exact scalar and finite conditional checks. Independent review and formalization remain pending. The documented baseline is the 2^{-293} coefficient derived in OpenAI's manuscript; no priority or strongest-known-bound claim is made.

1 Statement and dependencies

Proposed theorem 1. *For each known finite labeled matroid M on n elements there is a measurable, distribution-independent online rule using one sample per label and $38n+23$ initial independent fair bits with the following property. Let all nonnegative sample/value coordinates S_e, V_e be mutually independent, with matching laws at each label and an independent seed. If $\mathbb{E}\,\mathrm{OPT}_M(V) < \infty$, then for every measurable arrival permutation depending on all samples, values, and the entire seed, the rule maintains independence at every prefix and earns expectation at least $C_0\mathbb{E}\,\mathrm{OPT}_M(V)$, where*

$$C_0 = \frac{9652683394380719}{637507023635152349901619200} > 2^{-36}.$$

We use the density maps, guarded paths, hybrid safety comparison, marked-pivot encoding, rank accounting, and lossless simulation in OpenAI's source [1], pinned at commit `fd4aeeb2ee4f` (full revision in the linked source). We specify the rule and source interfaces here, prove the new quadratic transaction and exit bounds, and derive the refined residual and final constants. The structural matroid lemmas remain cited dependencies. No polynomial-time or bounded-precision claim is made.

2 The fixed-vector rule and its statistics

Fix a nonnegative hidden vector w before any seed bits. The rule reveals and sacrifices a seed-chosen mask and must handle every permutation; its target is $\mathbb{E}_R \min_\pi w(A(w, R, \pi))$. Write $\mathrm{r}(A \mid P) = \mathrm{r}(A \cup P) - \mathrm{r}(P)$ and $F_\varnothing = \mathrm{cl}(\varnothing)$. Fix a label order for every tie. Higher rounded weight means higher priority. Put $W = \mathrm{OPT}_M(u)$ and let I^* be priority greedy on the full rounded vector.

Use the parameters derived below:

$$\kappa = 2^{11}, \quad t = 2^{-16}, \quad \mathbb{P}(H_e) = 3/4, \quad \mathbb{P}(D_e) = 5/8, \quad \mathbb{P}(C_e) = \mathbb{P}(T_e) = t,$$

with main probability $1 - t$, fair parity, and uniform shared phase $J \in \{0, \dots, 63\}$. Put $\theta = 69/64$, $B = \theta^{64}$, and $u_J(x) = \theta^J B^{\lfloor \log_B(x/\theta^J) \rfloor}$ for $x > 0$, with $u_J(0) = 0$. All coins are independent and drawn initially.

The maximum branch reveals a separate fair mask and accepts the first unmasked positive nonloop with priority above every revealed positive nonloop. An empty threshold is below every positive nonloop. If m_* is the largest rounded nonloop weight, expected worst-order reward is at least $m_*/4$: the largest-priority nonloop is unmasked and the second is masked with probability $1/4$. With only one positive nonloop success has probability $1/2$; with none the bound is zero.

In the main branch reveal and sacrifice $H \cup D \cup C$. Define

$$U = \{e \notin H : u_e > 0,\ e \notin \mathrm{cl}(H_{>e})\}, \qquad Y = I^* \setminus H.$$

The candidate test uses only revealed H values and the tested value. For a fixed phase divide weights by θ^J, so true groups U_i have weight B^i. List those meeting D in increasing weight order as $G_1, \dots, G_m$; put $D_j = D \cap G_j$, and similarly C_j, T_j. The full groups are analysis objects; their revealed D_j, C_j parts suffice for preparation. If there are no listed groups, reject all.

For a flat P, let $Q_j(P)$ be the largest flat $Z \supseteq P$ maximizing $|D_j \cap Z| - \kappa\, \mathrm{r}(Z)$. Take closure first for nonflat arguments. Set $a_j = -3j$ and enable $Q_{j,k} = Q_j$ for $k \geq a_j$, with the identity earlier. Starting from $F_0(k) = F_\varnothing$ for $k < 0$ and E otherwise, define

$$X_j(k) = Q_{j,k}(F_{j-1}(k)), \qquad F_j(k) = \mathrm{cl}(X_j(k) \cup (C_j \cap X_j(k+1))).$$

Compute paths in increasing j on the finite relevant auxiliary timeline, and put $\widehat{F} = F_m$. For a candidate nonloop in listed group j, write $b_j(e) = \min\{k : e \in X_j(k)\}$. It is eligible precisely when it is unsacrificed, T-positive, has birth $b \geq a_j + 2$ of the chosen parity, and lies outside $\widehat{F}(b-2)$. At endpoint b maintain a set A_b; accept an eligible e exactly when $\mathrm{r}(\{e\} \mid \widehat{F}(b-2) \cup A_b) = 1$.

The source density and path lemmas give

$$F_{j-1}(k) \subseteq X_j(k) \subseteq F_j(k) \subseteq X_j(k+1), \qquad |D_j \cap (\mathrm{cl}(Q_j(P) \cup Z) \setminus Q_j(P))| \leq \kappa\, \mathrm{r}(Z \mid Q_j(P)).$$

Nonloops are born at a_j or in $[a_j + 2, 0]$, and the entire nominal path uses a generator set of size at most $|D_j|/\kappa$. The nested layer bases give prefix independence. Finite rank operations, fixed tie rules, and countably many rounding levels give measurability. These structural facts are inherited from the cited source.

For valid selected-parity endpoints $a_j + 2 \leq b \leq 0$, define

$$\begin{aligned}
K_{j,b} &= \bigcup_{\ell<j} (T_\ell \cap (X_\ell(b) \setminus X_\ell(b-1))), \\
P_{j,b} &= G_j \cap (X_j(b) \setminus X_j(b-1)), \quad O = D \cup C, \quad S_{j,b} = P_{j,b} \setminus O, \\
z_j &= \sum_b \mathrm{r}(S_{j,b} \mid X_j(b-2) \cup K_{j,b}), \\
\lambda_j &= \sum_b \mathrm{r}(T_j \cap S_{j,b} \mid \widehat{F}(b-2) \cup K_{j,b}).
\end{aligned}$$

The safe count σ_j counts $d \in D_j \cap Y$ with such a birth and $d \notin Q_j(F_{j-1}(b-2) \cup K_{j,b})$. For each nominal summand choose a fixed-order basis extension Z_b, so $\sum_b |Z_b| = z_j$. The residual is

$$\mathcal{R}_j = \bigcup_b (P_{j,b} \setminus \mathrm{cl}(X_j(b-2) \cup K_{j,b} \cup Z_b)) \subseteq O.$$

Extend σ_i, z_i, λ_i by zero on unlisted true groups. The source density argument gives $\sigma_j \le \kappa z_j + |\mathcal{R}_j|$. Its order-uniform lemma bounds the number of accepts of weight at least that of G_j below by λ_j, for every seed and order. One later-path generator budget over the whole timeline is

$$c_j = |C_j| + \sum_{\ell > j} (|C_\ell| + |D_\ell|/\kappa).$$

Conditional on H, D, C, parity, and test bits outside the focal group, independent test thinning gives $\mathbb{E}[\lambda_j \mid \mathcal{F}_{-j}] \ge t(z_j - c_j)$. These are the source interfaces used below.

3 A quadratic transaction inequality

Lemma 1. *Let $\mathcal{A}$ be an increasing event on finitely many independent bits, and let a predictable query procedure, reading each bit at most once, determine whether $\mathcal{A}$ occurs. Write $p = \mathbb{P}(\mathcal{A}^c)$. At the completed transcript let q be the failure probability with only unqueried bits random and every queried bit reset to zero. Then*

$$\mathbb{E}[\mathbf{1}_{\mathcal{A}} q^2] \ge p(1-p).$$

Proof. At an intermediate transcript, let W be its remaining coordinates and I its queried positives. Let m be the actual conditional failure probability, and q the failure probability with only W random and all other bits zero. Predictable product exposure identifies m with the probability when I is forced positive and W is random. Increasingness gives $0 \le m \le q \le 1$. Initially $m = q = p$.

For a predictably chosen next coordinate, the depleted probability q' is determined before the answer and satisfies $q' \ge q$. The actual conditional failure probability is a martingale: $\mathbb{E}[m' \mid \text{transcript}] = m$. Therefore the potential $\Phi(q, m) = q^2 - 2qm$ satisfies

$$\mathbb{E}[\Phi(q', m') \mid \text{transcript}] - \Phi(q, m) = (q' - q)(q' + q - 2m) \ge 0. \tag{1}$$

At the end $m = \mathbf{1}_{\mathcal{A}^c}$, and failure forces $q = 1$. The initial potential is $-p^2$; its final expectation is $\mathbb{E}[\mathbf{1}_{\mathcal{A}} q^2] - p$. Sum (1) over the finite number of queries, padding stopped paths by zeros, to obtain the result. The proof includes $p = 0, 1$ and transactions needing no queries. □

4 The reverse exit hazard

Fix the source's filter and density masks, a listed-group prefix, and a nonloop d still alive before a reverse step. Its current states are S, and its transition is $\mathcal{G}_k(c, S)$. The remaining guard coordinates have their product law. Let $\mathcal{A}$ be the increasing event that d survives, and let p be the conditional exit probability.

For completeness, for $k < 0$ the source reverse map on a listed-group prefix is defined recursively by

$$A_0 = F_\varnothing, \qquad N_j = Q_{j,k}(A_{j-1}), \qquad A_j = \mathrm{cl}(N_j \cup \{e \in G_j \cap S_j : c_e = 1\}),$$

with output $(N_j)_j$. Starting from $X_j(0) = E$, reverse steps compute $X(k) = \mathcal{G}_k(C, X(k+1))$. For actual later states, $N_j \subseteq S_j$ for every fresh bit assignment. After computing N_j, query only $G_j \cap (S_j \setminus N_j)$; omitted additions already lie in N_j. The new remaining domain is $\bigcup_j (G_j \cap N_j)$.

The source queries exactly the departing coordinates in predictable batches. Their answers determine all output states: unqueried bits are inside flats already spanned. Thus the transcript decides $\mathcal{A}$, on both survival and failure, so Lemma 1 applies.

Let S' be the actual output and V' the remaining coordinates. For each assignment c supported on V', monotonicity in time and state gives

$$\mathcal{G}_{k-1}(c, S') \subseteq \mathcal{G}_k(c, S).$$

The right side has all just-queried coordinates zero, exactly the depleted experiment defining q. Conditional on the full post-step transcript, the bits in V' remain independent. On survival the next hazard r therefore satisfies $r \geq q$. The states are frozen in this comparison; earlier states are not recomputed after resetting bits. No order is conditioned on.

The next alive squared hazard consequently obeys

$$\mathbb{E}[\mathbf{1}_{\text{survive}} r^2] - p^2 \geq p - 2p^2.$$

Telescope this potential over the deterministic finite reverse horizon. Every nonloop exits, and if P is its hazard at actual exit, then

$$-p_{\text{start}}^2 \geq \sum_i \mathbb{E}[\mathbf{1}_{\text{alive}} p_i] - 2\sum_i \mathbb{E}[\mathbf{1}_{\text{alive}} p_i^2] = 1 - 2\mathbb{E}P.$$

Hence

$$\boxed{\mathbb{E}P \geq (1 + p_{\text{start}}^2)/2 \geq 1/2.} \tag{2}$$

In particular, the source's total squared-exit mass is at least $1/2$. Its hybrid comparison still applies because guard and test rates agree. The same baseline-birth loss and fair parity give, after averaging a density mask of arbitrary rate ρ,

$$\mathbb{E}[\sigma_i \mid H] \geq \gamma |Y_i| - \frac{\rho n_i}{2\kappa}, \qquad \gamma = \rho/4. \tag{3}$$

This averages before conditioning on listing, extending the statistics by zero on unlisted groups.

The bound $1/2$ is sharp for the abstract hypotheses. Take N Bernoulli-t bits and repeatedly test the OR of the remaining suffix, querying until the first positive or until that suffix is exhausted. The next hazard equals the depleted current failure probability. With $a = 1 - t$, classification by the last positive (or none) gives

$$\mathbb{E}P = a^{2N} + t\sum_{j=0}^{N-1} a^{2j} = a^{2N} + \frac{1 - a^{2N}}{2 - t}.$$

For $t = 1/m$, $N = m^2$, this tends to $1/2$. This is an abstract limitation, not a hard matroid instance or a bound on the best prophet algorithm.

5 An integrated residual estimate

Fix H, parity, and all mask bits outside a true group of size $n \geq \kappa$; do not condition on its focal bits or listing. The residual family is fixed under this conditioning. The first marked sequence has at most $q = \lfloor n/\kappa \rfloor$ movable generators; the second also has at most q witnesses. The source encoding and marked-pivot lemma therefore give

$$|\mathfrak{R}| \leq 4^{3q} S_q(n)^3 S_{2q}(n) \leq \exp(An/\kappa), \qquad A = 5\log\kappa + 5 + 4\log 2.$$

This bound retains the floors as in the source binomial estimates.

Within the group the sacrificed mask has independent rate $p_O = \rho + (1-\rho)t$. Write $\lambda = -\log p_O > 0$. Let e_i be the actual residual size on the listed branch with $z_i \le n/\kappa$, and zero otherwise. For a family of size $N \ge 1$,

$$\mathbb{P}(e_i > x \mid \text{conditioning}) \le \min\{1, Ne^{-\lambda x}\} \quad (x \ge 0).$$

Integrating, then using $n \ge \kappa$, gives

$$\mathbb{E}[e_i \mid \text{conditioning}] \le \frac{\log N + 1}{\lambda} \le \frac{(A+1)n}{\kappa\lambda}.$$

Average away the parity and exterior bits to get the same final bound given H. An empty family gives zero error. The selected residual itself need not be independent of the mask. Since the source's $\sigma_i \le \kappa z_i + e_i$ holds on every branch, (3) implies

$$\mathbb{E}[z_i \mid H] \ge \frac{\gamma|Y_i| - \epsilon n_i}{\kappa}, \qquad \epsilon = \frac{\rho}{2\kappa} + \frac{A+1}{\kappa\lambda}. \tag{4}$$

6 Parameters and final accounting

Choose

$$\kappa = 2^{11}, \quad t = 2^{-16}, \quad h = \mathbb{P}(H_e) = 3/4, \quad \rho = \mathbb{P}(D_e) = 5/8, \quad \gamma = 5/32.$$

Guards and tests have rate t; parity is fair. Set main-branch probability $p_{\text{main}} = 1 - t$. The maximum branch retains its separate fair mask. For rounding take $\theta = 69/64$, $L = 64$, $B = \theta^L$, and uniform $J \in \{0, \dots, L-1\}$, with

$$u_J(x) = \theta^J B^{\lfloor \log_B(x/\theta^J) \rfloor}, \quad u_J(0) = 0.$$

The source rule otherwise stays the same. All choices are initial and independent.

The atanh series gives $\log 2 \le 25/36 < 7/10$ and $\log(8/5) > 1032/2197 > 15/32$. As $p_O = (5/8)(1 + 3t/5)$, an upper bound on the error in (4) is

$$\ell = 15/32 - 3t/5, \qquad \bar{\epsilon} = \frac{5}{16\kappa} + \frac{473}{10\kappa\ell} = \frac{248756209}{5033066496}.$$

Here $A + 1 = 59 \log 2 + 6 < 473/10$.

For general filter rate h, the elementary source filter argument gives

$$\mathbb{E}_H u(Y) = (1-h)W, \quad \mathbb{E}_H u(U) = \frac{1-h}{h}\mathbb{E}_H u(\text{greedy}(H)) \le \frac{1-h}{h}W.$$

Indeed, condition on preceding memberships in rounded priority: the span test is fixed, and the probabilities for a candidate and a greedy selected sample are respectively $1-h$ and h times that test. At $h = 3/4$ the two coefficients are $1/4$ and $1/3$.

Put $\beta = (B-1)/B$ and

$$L_0 = \bar{\epsilon} + t\kappa + \frac{t\kappa + \rho}{B-1}, \qquad s = (1-h)\left(\gamma - \frac{L_0}{h}\right).$$

The source's order-uniform rank comparison and independent test thinning give, for large groups,

$$\mathbb{E}[\lambda_i \mid H] \ge \frac{t}{\kappa}\left[\gamma|Y_i| - (\bar{\epsilon} + t\kappa)n_i - (t\kappa + \rho)\sum_{j>i} n_j\right].$$

The weighted higher-group sum is at most $u(U)/(B-1)$. Omitted groups $n_i < \kappa$ cost at most $\kappa m_*/\beta$ in comparison mass, where m_* is the maximum rounded nonloop weight. Consequently

$$\mathbb{E}_R \min_{\pi} \text{reward}_{\text{main}} \geq \frac{\beta t s}{\kappa} W - t\gamma m_*.$$

The maximum branch guarantees $m_*/4$ in worst-order expectation, and $(1 - p_{\text{main}})/4 = t/4 \geq p_{\text{main}} t\gamma$, cancelling the negative term.

For a positive x, let $k = \lfloor \log_\theta x \rfloor$. Its rounded values over all phases are θ^{k-r} for $r = 0, \ldots, L-1$, once each. Since $\theta^k \geq x/\theta$, the geometric sum gives

$$\mathbb{E}_J u_J(x) \geq \frac{\beta}{L(\theta - 1)} x = \frac{\beta}{5} x.$$

Apply this to a fixed true optimal independent set to obtain the same factor for expected rounded optimum. Thus the final coefficient is

$$C = \frac{p_{\text{main}} \beta^2 t s}{5\kappa}.$$

Exact integer arithmetic gives $69^{64} > 120 \cdot 64^{64}$, hence $\beta > 119/120$ and

$$s > \frac{\gamma}{4} - \frac{1}{3}\left(\bar{\epsilon} + t\kappa + \frac{t\kappa + \rho}{119}\right) = \frac{2652290047}{256686391296} > 0.$$

Substitution gives $C > C_0 > 2^{-36}$ as stated. Every comparison is certified with rational arithmetic. The exact construction coefficient is approximately $1.0459032\, 2^{-36}$; the smaller C_0 is simpler to report.

The minimum over all permutations remains inside the seed expectation at every stage. The source's glued-vector simulation therefore transfers this coefficient losslessly to one sample per label. Conditional on the entire seed the glued vector has the product value law; ignored samples, sacrificed values, and unused coins may all still inform the adversary's order.

7 Resources and evidence

Per label draw $2 + 3 + 16 + 16 + 1 = 38$ fair bits for the filter, density mask, guard, test, and maximum mask. Shared bits are $6 + 16 + 1 = 23$ for the phase, branch, and parity, including unused-branch randomness. All new scalar constants and phase multipliers are fixed rationals. Rank/closure tables and flat enumeration may be exponential and remain charged preprocessing and storage. Observation precision and integer level indices are unbounded in the original exact-real measurable model. Rational input implementations can compare exact rational powers, with costs depending on bit lengths and exponents. No polynomial complexity is claimed.

The local exact checks cover 4,916 reverse-process quadratic resource and drift inequalities and 3,153 global inequalities on 170 small matroid/weight instances. Additional checks cover all 199 increasing Boolean events on at most four bits at two rates, 5,992 biased-filter instances, 560 selected-residual expectations, and 40 rational-rounding boundary cases. The abstract sharpness formula is checked by enumeration. These are finite evidence; the universal claim rests on the written argument and cited source dependencies.

The same-agent audit rechecked the query martingale, predictable depleted probability, both terminal outcomes, frozen-state containment, telescope, integrated tail at $n = \kappa$, changed mask factors, rounding, and simulation. Independent review and formalization remain outstanding. Repository preparation does not constitute independent validation, a priority determination, or a confirmed best-known-bound claim.

8 Related work and development history

OpenAI's manuscript [1] derives a 2^{-293} coefficient in this model, stronger than its 2^{-310} headline. That derived coefficient is our documented baseline; the structural dependencies are identified above.

Daniel (@firsttimelifer) publicly announced a 2^{-47} guarantee on October 8, 2026 [2]. A supporting write-up was unavailable for comparison at the October 10 audit; we have not verified its assumptions or proof. We acknowledge it as an unverified announcement and do not use it as an established baseline.

This research effort progressed through 2^{-63}, 2^{-43}, and 2^{-36} on October 10, 2026. The first checkpoint refined source parameters, affine branch mixing, and finite rounding offsets. The second introduced a logarithmic hazard potential and tighter residual accounting. The present quadratic potential and integrated residual bound supersede those analyses. The dated archive preserves the intermediate proofs, certificates, and review status. These are stages of one research effort, not separate discoveries of the original constant-guarantee theorem.

References

[1] OpenAI, *One Sample Suffices for Matroid Prophet Inequalities against an Almighty Adversary*, September 23, 2026. Pinned manuscript index.

[2] Daniel (@firsttimelifer), Public announcement, October 8, 2026. Unverified; supporting proof unavailable at the audit date.